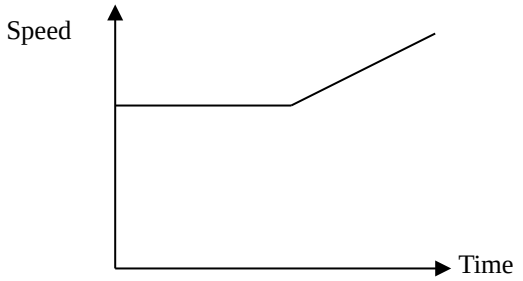
SCIENCE 1.6 **Name:** ***ANSWERS***

WORKSHEET FOUR **SPEED-TIME GRAPHS**

1. For each of the following speed-time graphs describe the journey that the graph represents.

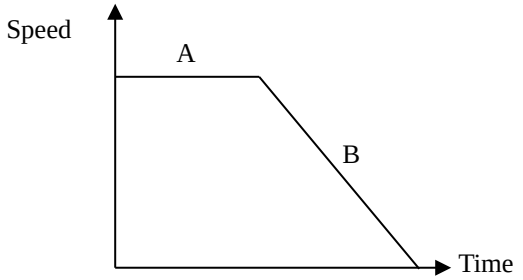
(a)



constant speed then

constant acceleration

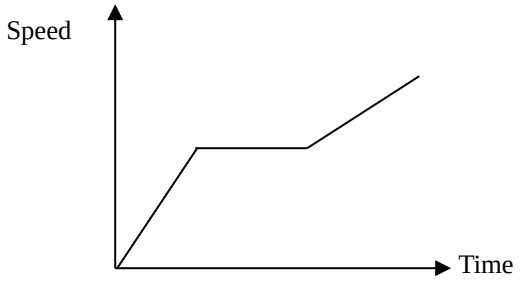
(b)



A ***constant speed***

B ***constant decelleration***

(c)



constant acceleration

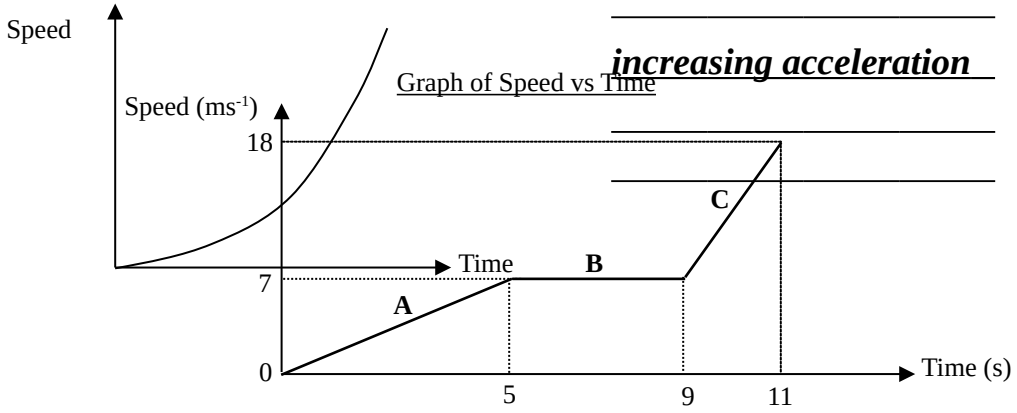
constant speed

constant acceleration

(d)

increasing acceleration

2.



(a) Describe the motion over parts A,B and C of the graph.

A ___***constant acceleration***_____

B ___***constant speed***_____

C ___***constant acceleration***_____

(b) Calculate the acceleration over parts A, and C of the graph.

A . $\boldsymbol{a} = {}^{rise}\!/_{run} = {}^{7}\!/_{5} =$ ***1.4 ms^{-2}***

C. $\boldsymbol{a} = {}^{rise}\!/_{run} = {}^{11}\!/_{2} =$ ***5.5 ms^{-2}***

(c) Calculate the distance travelled for each section of the journey and for the overall journey.

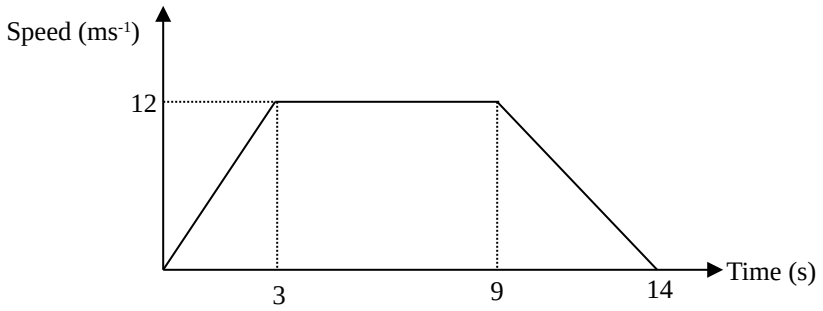
A. ${}^{35}\!/_{2} =$ **17.5 m**

B. 7 x 4 = **28 m**

C. 14 + 11 = **25 m**

Total distance = ___***70.5 m***_____

3.



(a) Describe the journey in your own words. (Give values if possible)

Constant acceleration of 4 ms^{-2}…. then constant speed of 12 ms^{-1}…. then a constant deceleration of 2.4 ms^{-2}

(b) Calculate the total distance covered in the journey.

Distance = 18 + 72 + 30 = 120 m